WiFi CSI-Based Long-Range Through-Wall Human Activity Recognition with the ESP32

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Abstract. WiFi Channel State Information (CSI)-based human activity recognition (HAR) is an unobtrusive method for contactless, long-range sensing in spatially constrained environments while preserving visual privacy. Despite the presence of numerous WiFi-enabled devices around us, few expose CSI to users, resulting in a lack of sensing hardware options. Recently, variants of the Espressif ESP32 have emerged as potential low-cost, easy-to-deploy solutions for WiFi CSI-based HAR. In this work, we evaluate the ESP32-S3's long-range through-wall HAR capabilities by combining it with a 2.4GHz directional biquad antenna. The experimental setup uses a transmitter-receiver configuration spanning 18.5m across five rooms. We assess line-of-sight (LOS) and non-line-of-sight (NLOS) performance using CNN HAR models trained on CSI spectrograms. CSI HAR datasets used in this work consist of 392 LOS and 384 NLOS spectrograms from three activity classes and are made publicly available. The gathered results clearly demonstrate the feasibility of long-range through-wall presence detection and activity recognition with the proposed setup.

Keywords: human activity recognition (HAR) · channel state information (CSI) · non-line-of-sight (NLOS) · through-wall sensing · ESP32

1 Introduction

WiFi-based Human Activity Recognition (HAR) exploits the measurable signal changes caused by human activities within a WiFi device's electromagnetic field to distinguish and recognize activities [10]. While camera-based approaches dominate the field, WiFi is gaining recognition as a sensing modality. WiFi is cost-effective, unobtrusive, illumination invariant, and protects the visual privacy of monitored individuals by not capturing any color or texture information, a key requirement in privacy-sensitive applications [2]. Additionally, WiFi

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signals can penetrate walls, enabling contactless long-range activity sensing in spatially constrained environments with operating ranges of up to 35m indoors [19]. This provides an economic advantage over camera-based approaches that require per-room deployment and opens up new possibilities like through-wall HAR [17], the focus of this work. Early WiFi-based HAR relied on the Received Signal Strength Indicator (RSSI) [18], which measures the signal strength of the WiFi channel at the receiver. Current approaches utilize Channel State Information (CSI), which captures amplitude and phase information of WiFi subcarriers [11]. CSI offers higher information density, enabling the recognition of fine-grained activities, and is more robust to environmental effects [12]. While all WiFi devices process CSI internally, most off-the-shelf devices do not expose this information to end users. Consequently, only specific hardware and software combinations allow CSI capture. Examples include the outdated Intel NIC 5300 with the Linux 802.11n CSI Tool[1] and Atheros NIC variants (AR9580, AR9590, AR9344, and QCA9558) with the Atheros CSI Tool[2]. Recently, the Nexmon CSI Tool[3] enabled CSI capture on Raspberry Pi [5]. Another emerging option is the Wi-ESP CSI Tool[4], which leverages the popular ESP32 microcontroller from Espressif Systems [3]. While a handful of works have explored the ESP32's potential in short-range line-of-sight (LOS) scenarios [6] and non-line-of-sight (NLOS) (through-wall) HAR scenarios [1,8], its activity sensing capabilities in long-range NLOS scenarios remain unexplored.

Contributions. In this work, we pair the ESP32-S3 with a 2.4GHz directional biquad antenna to enable long-range through-wall HAR in indoor environments. To the best of our knowledge, this is the first demonstration of such a setup for CSI-based through-wall HAR. We evaluate the proposed setup in long-range LOS and NLOS HAR scenarios, covering a distance of 18.5m across five rooms. A CSI HAR dataset containing 392 LOS and 384 NLOS samples of three activity classes is collected in an office environment and a performance evaluation on the basis of CNN HAR models trained on CSI spectrograms is conducted. The CSI HAR dataset used in this work is made publicly available.

2 Related Work

Through-wall CSI-based HAR has received significant attention, particularly with the Intel NIC 5300 being widely used as a sensing device, as highlighted in the comprehensive survey by Wang et al. [17]. However, due to its discontinuation in 2016, its sustainability for future CSI-based HAR applications is limited. Consequently, researchers have started exploring alternative solutions, such as Espressif ESP32 variants, which could offer a low-cost and easy-to-deploy solution to CSI-based HAR. While the ESP32 has been utilized in LOS scenarios

[1] Linux CSI Tool, https://dhalperi.github.io/, accessed: 05.06.2023.
[2] Atheros CSI Tool, https://wands.sg/, accessed: 05.06.2023.
[3] Nexmon CSI Tool, https://github.com/seemoo-lab/, accessed: 05.06.2023.
[4] Wi-ESP CSI Tool, https://wrlab.github.io/Wi-ESP/, accessed: 05.06.2023.

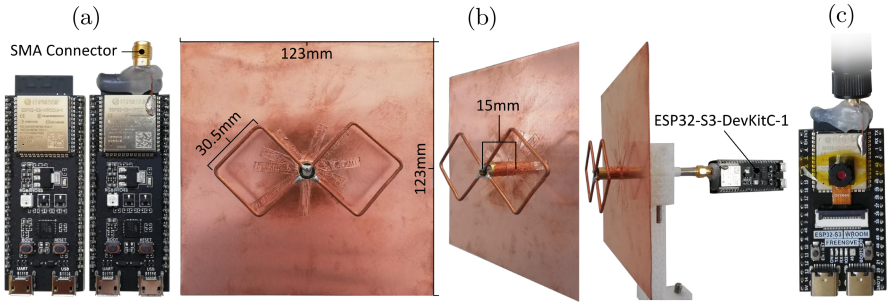


Fig. 1. (a) ESP32-S3-DevKitC-1 boards with built-in PCB antenna and SMA connector, (b) 2.4 GHz directional biquad antenna, and (c) ESP32-S3-WROOM-CAM board.

[3,6,7], there have been limited investigations into NLOS scenarios. To the best of our knowledge, only two works have explored NLOS scenarios to date [1,8]. In the work by Hernandez and Bulut [8], the feasibility of adversarial occupancy monitoring based on CSI is assessed. ESP32 devices are positioned as transmitters and receivers on the external wall of a hallway, successfully sensing the presence and walking direction of humans. Notably, the setup employs aluminum plates as RF shielding to improve signal quality by leveraging the built-in PCB antenna of the ESP32 as a crude directional antenna. Another work addressing NLOS scenarios is presented by Kumar et al. [1], where ESP32 devices are used for presence and fall detection. The conventional transmitter-receiver placement, with activities performed between the devices, is evaluated in a laboratory environment. Experimental results indicate the presence of characteristic CSI patterns induced by activities, even with up to two walls between the transmitter and receiver, demonstrating the feasibility of this approach.

3 Experimental Setup

We now describe the proposed setup, including its hardware components, the characteristics of the recording environment in which LOS and NLOS performance are evaluated, and the procedure for collecting CSI activity samples.

3.1 Hardware

Our hardware setup utilizes two ESP32-S3-DevKitC-1[5] boards equipped with the ESP32-S3-WROOM-1[6] module. The boards are connected using Espressif's ESP-NOW wireless communication protocol. In a transmitter-receiver configuration, CSI packets are sent at a fixed rate of 100 Hz and a Python script running on a notebook captures the incoming CSI packets from the receiver. While the ESP32-S3-WROOM-1 module has a built-in PCB antenna for basic WiFi

[5] ESP32-S3-DevKitC-1, https://docs.espressif.com/, accessed: 05.06.2023.
[6] ESP32-S3-WROOM-1, https://www.espressif.com/, accessed: 05.06.2023.

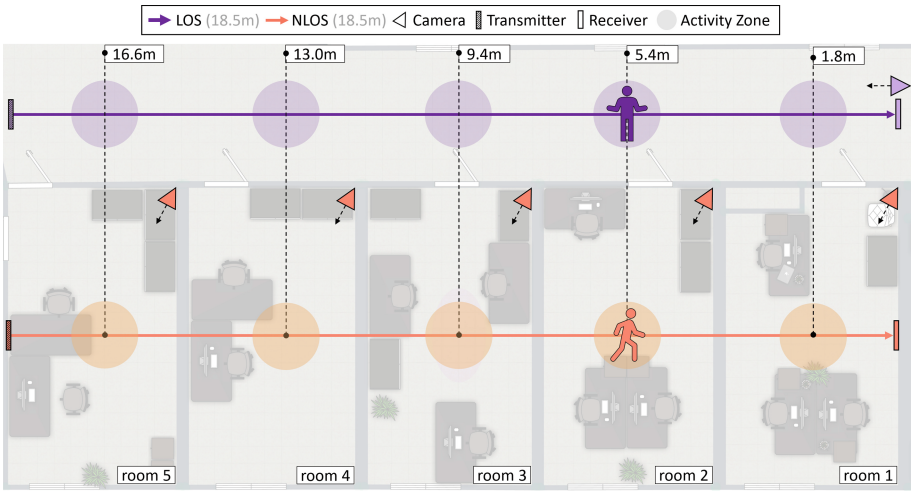


Fig. 2. Overview of the recording environment, showing the placement of hardware components for LOS and NLOS scenarios.

connectivity, we find it unsuitable for long-range through-wall HAR applications due to its low gain and unpredictability observed in preliminary short-range LOS tests. To address this, we replace the built-in PCB antenna with an SMA connector, enabling the use of an external antenna as depicted in Fig. 1a. We evaluate 2.4 GHz antennas, including a commercial omnidirectional dipole antenna with a gain of 8dBi, and a custom directional biquad antenna with a gain of 10–12dBi and a beamwidth of 70° [14], shown in Fig. 1b. To measure signal strength, we place the transmitter and receiver at the outer walls of the outermost rooms in the NLOS scenario (18.5 m across five rooms). The directional biquad antenna achieves the highest RSSI of –67 dB, outperforming the 8dBi omnidirectional dipole antenna with an RSSI of –79 dB. For reference, the built-in PCB antenna fails to establish a connection in this scenario. As a result of this test, we use a directional biquad antenna for both the transmitter and receiver, which not only provides a stable connection in long-range scenarios but also helps to eliminate external noise, which can be problematic with omnidirectional antennas [8]. Considering the relatively high RSSI of –67 dB, the proposed setup is likely to support even larger ranges and a higher through-wall count than evaluated in this work. Finally, for image-based activity labeling in NLOS scenarios, we incorporate an ESP32-S3-WROOM-CAM[7] (camera) board as shown in Fig. 1c, streaming images to the notebook.

3.2 Recording Environment

The evaluation of the proposed setup takes place in an office environment depicted in Fig. 2. It consists of a hallway measuring 18.5 m in length, with

[7] ESP32-S3-WROOM-CAM, https://github.com/Freenove/, accessed: 05.06.2023.

five adjacent rooms containing office furniture. These rooms, separated by 25 cm thick brick walls, present a challenging long-range NLOS scenario. The uniform size of the rooms (approximately $3.5\,\text{m} \times 6.0\,\text{m}$) and their arrangement allow for a direct comparison of LOS and NLOS HAR performance at various distances between the transmitter and receiver. For the LOS scenario (purple line in Fig. 2), the transmitter and receiver are positioned at opposite ends of the hallway, facing each other. To capture activity images, the camera is placed next to the transmitter, aligned in its direction. In the NLOS scenario (orange line), the transmitter and receiver are again facing each other, but positioned at the outer walls of rooms 5 and 1, respectively. The alignment of antennas is achieved by fine-adjusting the receiver's horizontal position based on the RSSI. Additionally, to capture activity images in the NLOS scenario, the camera is placed in the room where the activity occurs. The transmitter-receiver spacing remains constant at 18.5 m for both LOS and NLOS scenarios.

3.3 Data Collection and Pre-processing

To assess the feasibility of long-range through-wall HAR with the proposed setup, we collect CSI activity samples in the recording environment. The goal is to detect a person's presence and distinguish between coarse and fine body movements (e.g., walking vs. arm movements). Activities are performed within five circular zones (1.5 m radius) along the LOS and NLOS paths shown in Fig. 2. These zones are located at distances of $\{1.8, 5.4, 9.4, 13.0, 16.6\}$m from the receiver, corresponding to room centers in the NLOS scenario. For both scenarios, we record two minutes of continuous "walking" and "walking + arm-waving" activities in each zone, as well as five minutes of "no activity." Pre-processing involves cleaning the raw CSI recordings using images from the camera and outlier removal with the Hampel filter [13]. The filtered CSI time series data is then converted into time-frequency plots of subcarrier amplitudes over time, i.e. spectrograms. For this, recordings are divided into segments of 400 CSI packets (4-s time intervals @100 Hz sending frequency) and the amplitudes of 52 L-LTF subcarriers are plotted, resulting in a spectrogram size of 400×52. Examples of spectrograms for each activity class are shown in Fig. 3. To evaluate presence detection and activity recognition, the collected spectrograms are labeled. Presence detection labels $\{0, 1, 2, 3, 4, 5\}$ (0 for "no activity") correspond to room numbers, resulting in DP_{LOS} and DP_{NLOS}. Activity recognition labels $\{0, 1, 2\}$ correspond to depicted activities, resulting in DA_{LOS} and DA_{NLOS}. For the evaluation, a 8:1:1 training-validation-test split is used for all datasets. Dataset characteristics are summarized in Table 1. Furthermore, all datasets used in this work are publicly available[8].

[8] CSI HAR datasets, https://zenodo.org/record/8021099, accessed: 19.07.2023.

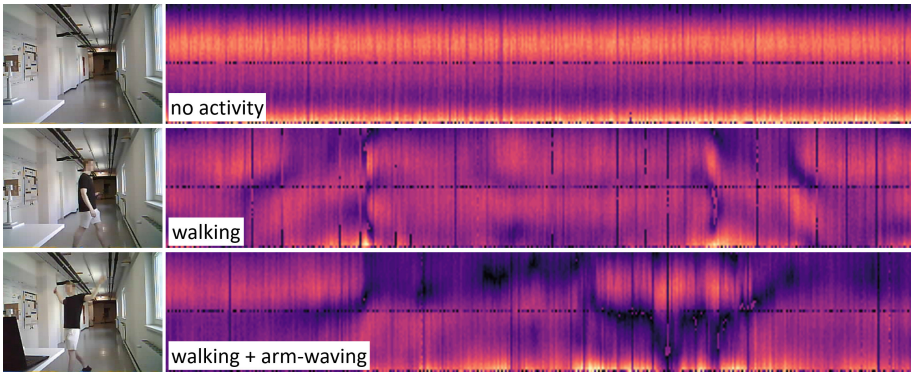


Fig. 3. LOS activity spectrograms of "no activity", "walking", and "walking + arm-waving", captured at a distance of 1.8 m.

Table 1. Characteristics of presence detection and activity recognition datasets.

Dataset	Scenario	Rooms	Persons	Classes	CSI Sampling	Interval	Spectrograms
DP_{LOS}	LOS	1	1	6 rooms	100 Hz	4 s	392
DP_{NLOS}	NLOS	5	1	6 rooms	100 Hz	4 s	384
DA_{LOS}	LOS	1	1	3 activities	100 Hz	4 s	392
DA_{NLOS}	NLOS	5	1	3 activities	100 Hz	4 s	384

4 Evaluation

As shown in [4], CSI spectrograms can be processed using Convolutional Neural Networks (CNNs) for person-centric sensing applications. Following this approach, we train presence detection and activity recognition models on the collected CSI HAR datasets. Separate LOS and NLOS models are trained for both problems to assess the impact of environmental effects, allowing direct comparisons. Generalization between scenarios is explored by training models in one scenario and testing in the other.

4.1 Model Training

To ensure reproducibility, we train presence detection and activity recognition models using the standard *torchvision.models* implementation of the EfficientNetV2 small architecture [15]. Presence detection models, namely P_{LOS} and P_{NLOS}, are trained on DP_{LOS} and DP_{NLOS}, respectively. Additionally, a presence detection model for both scenarios, $P_{LOSNLOS}$, is trained on $DP_{LOS} \cup DP_{NLOS}$. The same approach is used for activity recognition, resulting in models A_{LOS}, A_{NLOS}, and $A_{LOSNLOS}$. All models are trained from scratch and undergo 1k epochs of training using the Adam optimizer with a learning rate of 0.0001 and a batch size of 16. To eliminate class imbalances in the training

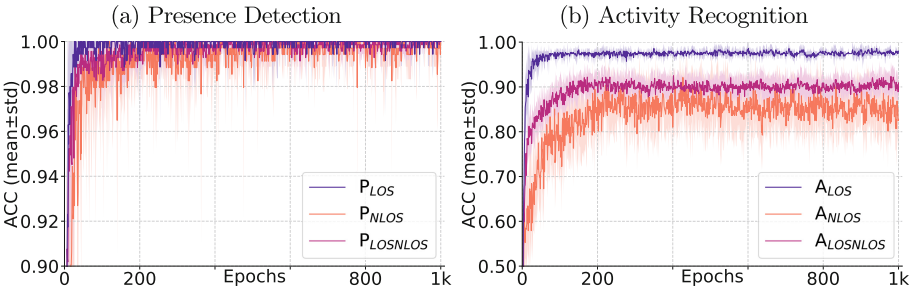


Fig. 4. Validation accuracy (mean±std across ten runs) of (a) presence detection and (b) activity recognition models over 1k training epochs.

Table 2. Performance of presence detection models P_{LOS}, P_{NLOS} and $P_{LOSNLOS}$, reported as mean and standard deviation across ten independent training runs.

Model	Test Dataset	Precision	Recall	F1 Score	ACC
P_{LOS}	DP_{LOS}	98.67 ± 1.8	98.61 ± 1.9	98.64 ± 1.8	98.46 ± 2.1
P_{NLOS}	DP_{NLOS}	98.14 ± 1.6	97.72 ± 2.2	97.93 ± 1.9	97.89 ± 2.0
P_{LOS}	DP_{NLOS}	15.79 ± 8.1	18.49 ± 7.3	16.61 ± 7.1	17.89 ± 7.1
P_{NLOS}	DP_{LOS}	15.71 ± 9.6	21.67 ± 4.5	17.29 ± 7.7	23.59 ± 5.6
$P_{LOSNLOS}$	DP_{LOS}	97.13 ± 2.8	96.57 ± 3.4	96.85 ± 3.1	96.41 ± 3.7
$P_{LOSNLOS}$	DP_{NLOS}	94.71 ± 2.9	91.11 ± 5.7	92.85 ± 4.3	92.37 ± 4.9
$P_{LOSNLOS}$	$DP_{LOS} \cup DP_{NLOS}$	98.33 ± 1.2	98.26 ± 1.6	98.29 ± 1.2	98.31 ± 1.2

dataset, a balanced sampler is employed. Data augmentation includes random circular shifts along the time axis applied to the spectrograms. For each model, ten independent training runs are conducted, and the mean and standard deviation of metrics such as precision, recall, F1-score, and classification accuracy are reported based on the respective test datasets.

4.2 Presence Detection Results

Figure 4a visualizes the training progress of presence detection models P_{LOS} and P_{NLOS}. In the LOS scenario, rapid convergence is observed, with the highest validation performance reached after just 26 epochs. Convergence in the NLOS scenario takes longer, with 118 epochs required. Table 2 presents detailed performance metrics on the corresponding test datasets, DP_{LOS} and DP_{NLOS}. Comparable presence detection accuracies are achieved in both scenarios, with P_{LOS} slightly outperforming P_{NLOS} (98.46 ± 2.1 vs. 97.89 ± 2.0), clearly demonstrating the feasibility of long-range through-wall presence detection with the proposed setup. To explore model generalization between scenarios, we evaluate the performance of P_{LOS} and P_{NLOS} on each other's test datasets, as shown in Table 2. A significant drop in performance, close to random guess-

Table 3. Performance of activity recognition models A_{LOS}, A_{NLOS} and $A_{LOSNLOS}$ reported as mean and standard deviation across ten independent training runs.

Model	Test Dataset	Precision	Recall	F1 Score	ACC
A_{LOS}	DA_{LOS}	97.80 ± 0.9	98.04 ± 0.9	97.92 ± 0.9	97.43 ± 1.1
A_{NLOS}	DA_{NLOS}	90.94 ± 3.8	87.92 ± 4.3	89.39 ± 4.0	88.16 ± 4.6
A_{LOS}	DA_{NLOS}	16.77 ± 5.4	33.60 ± 1.3	22.03 ± 4.5	42.10 ± 1.2
A_{NLOS}	DA_{LOS}	25.73 ± 10.6	32.90 ± 1.5	28.07 ± 5.6	41.79 ± 1.6
$A_{LOSNLOS}$	DA_{LOS}	96.17 ± 2.1	95.98 ± 2.3	96.07 ± 2.2	95.13 ± 2.7
$A_{LOSNLOS}$	DA_{NLOS}	80.35 ± 3.5	78.24 ± 6.2	79.23 ± 4.8	75.53 ± 5.3
$A_{LOSNLOS}$	$DA_{LOS} \cup DA_{NLOS}$	93.42 ± 1.3	92.16 ± 1.2	92.78 ± 1.2	91.95 ± 1.5

ing, is observed for both models. This outcome is expected, as poor generalization to entirely new environments is a well-known problem in WiFi-based HAR [9,16,18]. However, the model trained on both scenarios, $P_{LOSNLOS}$, achieves presence detection accuracies of 96.41 ± 3.7 and 92.37 ± 4.9 on the LOS and NLOS test datasets, respectively. Furthermore, a presence detection accuracy of 98.31 ± 1.2 is achieved on the combined test dataset, demonstrating generalization between LOS and NLOS scenarios.

4.3 Activity Recognition Results

Figure 4b visualizes the training progress of activity recognition models A_{LOS} and A_{NLOS}, highlighting the increased difficulty of the problem. Convergence requires a significantly higher number of training epochs compared to the presence detection problem. A_{LOS} and A_{NLOS} reach their peak validation performance after 349 and 426 epochs, respectively. Detailed performance metrics on the corresponding test datasets, DA_{LOS} and DA_{NLOS}, are provided in Table 3. Unlike presence detection, a larger performance gap between LOS and NLOS scenarios is observed (97.43 ± 1.1 vs. 88.16 ± 4.6). Interestingly, in the NLOS scenario, there is a relatively high standard deviation of 4.6% points, indicating significant run-to-run variance, with individual models occasionally achieving LOS-level performance. These promising results, obtained with a standard CNN architecture and minimal data augmentation, clearly demonstrate the feasibility of WiFi CSI-based long-range through-wall HAR using the proposed setup. As reported in Table 3, we also observe poor model generalization between scenarios in the activity recognition problem. However, both A_{LOS} and A_{NLOS} models achieve activity classification accuracies above random guessing (42.10 ± 1.2 and 41.79 ± 1.6), indicating some level of generalization. Furthermore, the model trained on both scenarios, $A_{LOSNLOS}$, achieves an activity classification accuracy of 91.95 ± 1.5 on the combined test dataset. Interestingly, we see overfitting on the LOS scenario, evidenced by the unbalanced classification accuracies in the individual scenarios (95.13 ± 2.7 and 75.53 ± 5.3). This discrepancy can be attributed to the relative ease of fitting the LOS data, as indicated by the previous results.

5 Conclusion

In this work, we proposed a novel setup for CSI-based HAR, combining the ESP32-S3 with a 2.4 GHz directional biquad antenna, which enables long-range through-wall HAR in spatially constrained environments. To assess performance in LOS and NLOS scenarios, we deployed the setup in a transmitter-receiver configuration within a realistic office environment spanning 18.5 m across five rooms. We collected a dataset of 392 LOS and 384 NLOS activity spectrograms, which is made publicly available. Activity recognition models based on the EfficientNetV2 small architecture, trained on this dataset, achieved accuracies of 97.43 ± 1.1 and 88.16 ± 4.6 in LOS and NLOS scenarios, respectively, clearly demonstrating the potential of the proposed setup for long-range through-wall CSI-based HAR applications.

References

1. Ajit Kumar, S., Akhil, K., Udgata, S.K.: Wi-Fi signal-based through-wall sensing for human presence and fall detection using ESP32 module. In: Udgata, S.K., Sethi, S., Gao, X.Z. (eds.) Intelligent Systems. LNNS, vol. 431, pp. 459–470. Springer, Singapore (2023). https://doi.org/10.1007/978-981-19-0901-6_41
2. Arning, K., Ziefle, M.: Get that camera out of my house! Conjoint measurement of preferences for video-based healthcare monitoring systems in private and public places. In: Geissbühler, A., Demongeot, J., Mokhtari, M., Abdulrazak, B., Aloulou, H. (eds.) ICOST 2015. LNCS, vol. 9102, pp. 152–164. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-19312-0_13
3. Atif, M., Muralidharan, S., Ko, H., Yoo, B.: Wi-ESP–a tool for CSI-based device-free Wi-Fi sensing (DFWS). J. Comput. Des. Eng. **7**(5), 644–656 (2020). https://doi.org/10.1093/jcde/qwaa048
4. Gao, Q., Wang, J., Ma, X., Xueyan, F., Wang, H.: CSI-based device-free wireless localization and activity recognition using radio image features. IEEE Trans. Veh. Technol. 1 (2017). https://doi.org/10.1109/TVT.2017.2737553
5. Gringoli, F., Schulz, M., Link, J., Hollick, M.: Free your CSI: a channel state information extraction platform for modern Wi-Fi chipsets. In: Proceedings of the 13th International Workshop on Wireless Network Testbeds, Experimental Evaluation & Characterization, pp. 21–28. WiNTECH '19, Association for Computing Machinery, New York, NY, USA (2019). https://doi.org/10.1145/3349623.3355477
6. Hao, Z., Wang, G., Dang, X.: Car-sense: vehicle occupant legacy hazard detection method based on DFWs. Appl. Sci. **12**, 11809 (2022). https://doi.org/10.3390/app122211809
7. Hernandez, S.M., Bulut, E.: Performing Wi-Fi sensing with off-the-shelf smartphones. In: 2020 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops), pp. 1–3 (2020). https://doi.org/10.1109/PerComWorkshops48775.2020.9156194
8. Hernandez, S.M., Bulut, E.: Adversarial occupancy monitoring using one-sided through-wall WiFi sensing. In: ICC 2021 - IEEE International Conference on Communications, pp. 1–6 (2021). https://doi.org/10.1109/ICC42927.2021.9500267
9. Kosba, A.E., Saeed, A., Youssef, M.: RASID: a robust WLAN device-free passive motion detection system. In: 2012 IEEE International Conference on Pervasive

Computing and Communications, pp. 180–189 (2012). https://doi.org/10.1109/PerCom.2012.6199865
10. Liu, J., Liu, H., Chen, Y., Wang, Y., Wang, C.: Wireless sensing for human activity: a survey. IEEE Commun. Surv. Tutor. **22**(3), 1629–1645 (2020). https://doi.org/10.1109/COMST.2019.2934489
11. Palipana, S., Rojas, D., Agrawal, P., Pesch, D.: Falldefi: ubiquitous fall detection using commodity wi-fi devices. PACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT) **1** (2018). https://doi.org/10.1145/3161183
12. Parameswaran, A.T., Husain, M.I., Upadhyaya, S., et al.: Is RSSI a reliable parameter in sensor localization algorithms: an experimental study. In: Field Failure Data Analysis Workshop (F2DA09), vol. 5. IEEE Niagara Falls, NY, USA (2009)
13. Pearson, R.K., Neuvo, Y., Astola, J., Gabbouj, M.: Generalized Hampel filters. EURASIP J. Adv. Signal Process. **2016**(1), 1–18 (2016). https://doi.org/10.1186/s13634-016-0383-6
14. Singh, B., Singh, A.: A novel biquad antenna for 2.4 GHZ wireless link application: a proposed design. Int. J. Electron. Commun. Technol. **3**(1), 174–176 (2012)
15. Tan, M., Le, Q.V.: Efficientnetv2: smaller models and faster training (2021)
16. Wang, W., Liu, A.X., Shahzad, M., Ling, K., Lu, S.: Device-free human activity recognition using commercial Wi-Fi devices. IEEE J. Sel. Areas Commun. **35**(5), 1118–1131 (2017). https://doi.org/10.1109/JSAC.2017.2679658
17. Wang, Z., Jiang, K., Hou, Y., Huang, Z., Dou, W., Zhang, C., Guo, Y.: A survey on CSI-based human behavior recognition in through-the-wall scenario. IEEE Access 1 (2019). https://doi.org/10.1109/ACCESS.2019.2922244
18. Youssef, M., Mah, M., Agrawala, A.: Challenges: device-free passive localization for wireless environments. In: Proceedings of the 13th Annual ACM International Conference on Mobile Computing and Networking, pp. 222–229 (2007)
19. Zafari, F., Gkelias, A., Leung, K.: A survey of indoor localization systems and technologies. IEEE Commun. Surv. Tutor. (2017). https://doi.org/10.1109/COMST.2019.2911558